

- 10 A polystyrene cup contains a mass of 88 g of water at 30 °C. A cube of ice of mass 12 g and temperature 0 °C is placed in the water. The water, of specific heat capacity $4.2 \times 10^3 \text{ J kg K}^{-1}$, is stirred till all the ice melts.

Assuming negligible heat loss to the cup and surroundings, what is the final temperature of the water?

specific latent heat of fusion of ice = $3.3 \times 10^5 \text{ J kg}^{-1}$.

A 17 °C

B 19 °C

C 21 °C

D 23 °C

A heat gained by ice in melting + heat gained by melted ice water = heat lost by water

$$12 [3.3 \times 10^5 + 4.2 \times 10^3 \times (T - 0)] = 88 \times 4.2 \times 10^3 \times (30 - T)$$

$$3.96 \times 10^3 + 50.4 T = 11.088 \times 10^3 - 369.6 T$$

$$420 T = 7128$$

$$T = 17 \text{ °C}$$

distractor: $T = 19 \text{ °C}$ (melted ice omitted)

11 An ideal gas undergoes a cycle of changes $A \rightarrow B \rightarrow C \rightarrow A$ as shown